Design and Implementation of a Web-Based LPG Distribution Management Information System with Multi-Tenant Architecture

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ARTICLE INFO	**ABSTRACT**
Article history:	**Background:** Liquefied Petroleum Gas (LPG) distribution is a critical component of Indonesia's national energy supply chain, with consumption exceeding 8.2 million tons annually. However, many LPG agents continue to rely on manual recording methods, leading to data inaccuracies, order tracking difficulties, and delayed report generation. **Objective:** This study aims to design and implement a web-based LPG distribution management information system utilizing multi-tenant architecture to address operational inefficiencies in LPG agent businesses. **Methods:** The system was developed using the Waterfall methodology with Unified Modeling Language (UML) for system design. The technology stack comprises React 18 for the frontend, NestJS 10 for the backend REST API, and PostgreSQL 15 for database management. Prisma ORM was employed for type-safe database access. Black-box testing was conducted to validate system functionality. **Results:** The developed system, named SIM4LON, successfully integrates order management, stock tracking, payment processing, and automated reporting within a unified platform. All 28 test cases passed with a 100% success rate, confirming that all functional requirements were met. The multi-tenant architecture effectively isolates data between different retail outlets (pangkalan) using row-level filtering based on tenant identifiers. **Conclusion:** The SIM4LON system successfully addresses the identified operational challenges faced by LPG agents. The implementation of modern web technologies combined with multi-tenant architecture provides a scalable, maintainable solution that improves operational efficiency and data accuracy in LPG distribution management.
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1. Introduction

Liquefied Petroleum Gas (LPG) represents one of the most strategically important energy sources widely utilized by Indonesian households and commercial establishments. According to the Ministry of Energy and Mineral Resources (2024), national LPG consumption has reached over 8.2 million tons annually, with an average growth rate of 4-5% each year [1]. This increasing consumption has driven greater complexity in managing LPG distribution from producers to end consumers.

Indonesia's LPG distribution chain involves several interconnected stakeholders. PT Pertamina, as the producer, supplies LPG to SPBE (Bulk LPG Filling Stations), then agents as authorized distributors collect supplies from SPBE for distribution to retail outlets (pangkalan), which finally sell directly to end consumers [2]. A typical agent manages between tens to hundreds of retail outlets with significant daily transaction volumes, encompassing ordering, delivery, payment, and reporting processes that require systematic management.

Based on observations and interviews conducted with multiple LPG agents in the Cianjur Regency area during October-November 2025, it was discovered that the majority of agents still employ manual recording systems using handwritten ledgers, paper receipts, and unintegrated Excel spreadsheets. This condition creates several significant operational problems. First, stock data inaccuracy frequently occurs because manual recording is prone to input errors and delayed updates, with discrepancies between physical and recorded stock reaching 5-10% monthly. Second, order tracking difficulties make it challenging for operators to monitor delivery status in real-time, resulting in inefficient coordination with drivers. Third, slow report generation causes monthly recapitulation processes to require 3-5 working days. Fourth, data loss risk remains high because paper-based records are susceptible to damage, loss, or illegibility.

Several previous studies have addressed LPG distribution information systems. Harahap et al. (2023) developed a web-based LPG 3 kg sales information system at PT. Nafa Energi Indonesia using PHP and MySQL [3]. Kurniawan and Saputra (2022) designed an LPG sales information system for retail outlets using the Waterfall methodology [4]. Rahman et al. (2024) implemented distribution requirement planning methods for gas inventory optimization [5]. Furthermore, Hidayat and Santoso (2023) created a web-based LPG distribution management system using the Laravel framework [6]. However, these studies have not implemented multi-tenant architecture that enables a single system to serve multiple retail outlets with strict data isolation.

Multi-tenant architecture has emerged as a fundamental design pattern in modern Software-as-a-Service (SaaS) applications, allowing multiple customers (tenants) to share a single application instance while maintaining complete data isolation [7]. Recent research by Kumar et al. (2024) demonstrates that integrating modern technologies such as IoT and digital platforms can significantly enhance LPG distribution operational efficiency by up to 30% [8]. Additionally, the adoption of cloud-native technologies for multi-tenant SaaS applications has been identified as a best practice for achieving scalability and cost efficiency [9].

This research addresses the identified gap by designing and implementing a web-based LPG distribution management information system called SIM4LON (Online LPG Distribution Management Information System) with multi-tenant architecture. The system employs modern web technologies including React for the frontend user interface, NestJS for the backend API services, and PostgreSQL for data persistence [10][11][12]. The multi-tenant approach uses a shared database with shared schema strategy, utilizing `pangkalan_id` as the tenant identifier to ensure data isolation while optimizing resource utilization.

The research objectives are: (1) to design an LPG distribution management information system using UML modeling that accommodates multi-tenant architecture; (2) to implement the system using React 18, NestJS 10, Prisma ORM, and PostgreSQL 15; and (3) to validate system functionality through comprehensive black-box testing.

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2. Method

2.1 Type and Approach of Research

This research employs a Research and Development (R&D) approach with the Waterfall software development methodology. The Waterfall model was selected because all system requirements were clearly identified through field observations and stakeholder interviews [13]. This sequential approach ensures systematic progression through requirements analysis, system design, implementation, testing, and deployment phases, providing traceability between requirements and implemented features.

2.2 Object and Scope of Research

The research object is the LPG distribution business process at medium-scale agents managing 10-100 retail outlets (pangkalan) in the Cianjur Regency, West Java, Indonesia. The scope encompasses the complete distribution workflow including order management, stock tracking, payment processing, and reporting. The system boundary excludes integration with Pertamina's centralized systems, which remains a recommendation for future development.

2.3 Data Collection Techniques

Three data collection techniques were employed in this research:

Observation: Direct observation was conducted over two weeks at LPG agents in Cianjur Regency during October-November 2025. Observed aspects included order placement workflows from retail outlets to agents, recording and administrative processes, delivery and distribution mechanisms, and payment and billing procedures.

Interviews: Semi-structured interviews were conducted with 10 stakeholders comprising 2 LPG agent owners, 3 agent operators/administrators, and 5 retail outlet owners. Interview questions focused on daily operational challenges, desired system features, and expectations for a new computerized system.

Literature Review: Academic literature was reviewed covering management information systems theory, LPG distribution in Indonesia, multi-tenant architecture design patterns, modern web development technologies, UML modeling standards, and the Waterfall development methodology.

2.4 Tools and Materials Used

The development environment utilized the following technology stack as shown in Table 1.

Table 1. Software Development Tools

Category	Tool	Version	Function
IDE	Visual Studio Code	1.85	Code editor
Frontend Framework	React	18.2	User interface library
Build Tool	Vite	5.0	Frontend bundler and dev server
CSS Framework	Tailwind CSS	3.4	Utility-first styling
UI Components	Shadcn/UI	Latest	Accessible component library
Backend Framework	NestJS	10.2	REST API framework
ORM	Prisma	5.7	Type-safe database access
Database	PostgreSQL	15	Relational data storage
UML Tool	PlantUML	Latest	UML diagram generation
Frontend Hosting	Vercel	-	Static site deployment
Backend Hosting	Railway	-	Container deployment
File Storage	Supabase Storage	-	Cloud file management

2.5 Research Procedures or Stages

The research followed the Waterfall methodology stages:

Stage 1 - Requirements Analysis (October 2025): Conducted stakeholder interviews and field observations to identify functional and non-functional requirements. Documented requirements in structured format with acceptance criteria.

Stage 2 - System Design (November 2025): Created comprehensive UML documentation including Use Case Diagrams (18 use cases, 3 actors), Class Diagrams (22 classes, 9 enumerations), Activity Diagrams (25 diagrams for core processes), Sequence Diagrams (18 diagrams for use case interactions), State Machine Diagrams (2 diagrams for order and payment status), and Entity Relationship Diagrams (11 tables). Additionally, designed the user interface wireframes and system architecture.

Stage 3 - Implementation (November-December 2025): Developed the system following the designed specifications. Frontend development used React with TypeScript, implementing responsive design patterns. Backend development utilized NestJS with modular architecture, implementing RESTful APIs with JWT authentication. Database implementation used PostgreSQL with Prisma for migrations and queries.

Stage 4 - Testing (December 2025): Conducted black-box testing with 28 test cases covering all functional requirements. Documented test results and performed bug fixes as necessary.

Stage 5 - Deployment (December 2025): Deployed the frontend to Vercel and backend to Railway with PostgreSQL database. Configured production environment variables and verified system accessibility.

![INSERT FIGURE: Waterfall Methodology Stages Diagram]

Figure 1. Waterfall Methodology Implementation Stages

2.6 Data Analysis Techniques

System validation was performed using black-box testing methodology, which focuses on testing system functionality without examining internal code structure [14]. Test cases were designed based on functional requirements, with each test case specifying input data, expected output, and pass/fail criteria. The test coverage included authentication (5 tests), order management (8 tests), payment processing (5 tests), stock management (6 tests), and reporting (4 tests).

The success rate was calculated using the formula:

Success Rate (%) = (Number of Passed Tests / Total Test Cases) × 100

A minimum success rate of 100% was required before deployment to ensure all functional requirements were met.

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3. Results and Discussion

3.1 Presentation of Research Results

3.1.1 Current System Analysis

The existing LPG agent business process operates entirely through manual methods. Retail outlets place orders via WhatsApp messages or phone calls to operators. Operators manually record orders in handwritten ledgers, then verbally assign drivers for delivery. Payments are separately recorded in cash books, and monthly reports are compiled manually using Excel spreadsheets. This workflow is illustrated in Figure 2.

![INSERT FIGURE: Current System BPMN Diagram]

Figure 2. Business Process Model of Current Manual System

The analysis identified the following critical issues: (1) no integration between order recording, delivery tracking, and payment systems; (2) high susceptibility to human error in manual data entry; (3) inability to monitor delivery status in real-time; (4) time-consuming monthly report generation requiring 3-5 working days.

3.1.2 Requirements Specification

Functional Requirements:

- FR-01: The system shall manage user accounts with role-based access control (Admin, Operator)
- FR-02: The system shall manage retail outlet (pangkalan) and driver master data

- FR-03: The system shall create, update, and track order status through lifecycle
- FR-04: The system shall record payments including down payments, installments, and full payments
- FR-05: The system shall track stock movements (incoming from SPBE, outgoing to retail outlets)
- FR-06: The system shall generate reports in PDF and Excel formats

Non-Functional Requirements:

- NFR-01: Web-based responsive design accessible on desktop and mobile browsers
- NFR-02: JWT-based authentication with single-session login enforcement
- NFR-03: Support minimum 100 retail outlets per agent
- NFR-04: Response time under 3 seconds for standard operations

3.1.3 System Design

Use Case Diagram: Figure 3 presents the complete use case diagram showing 18 identified use cases and 3 actor types. The Admin actor has full system access, the Operator handles daily operations, and the Pangkalan (retail outlet) actor has limited access to their own data through the multi-tenant isolation mechanism.

![INSERT FIGURE: Use Case Diagram]

Figure 3. SIM4LON Use Case Diagram

The 18 identified use cases are organized by actor role. Admin-exclusive use cases include user management and product configuration. Operator use cases encompass order creation and tracking, payment recording, stock management, and report generation. Pangkalan use cases enable retail outlets to view their orders and payments, and manage their consumer transactions within the multi-tenant data isolation framework.

Entity Relationship Diagram: Figure 4 displays the database schema comprising 11 core tables implementing the multi-tenant data model.

![INSERT FIGURE: Entity Relationship Diagram]

Figure 4. Entity Relationship Diagram

The database design implements the shared database, shared schema multi-tenant pattern. Data isolation is enforced through `pangkalan_id` foreign keys on transaction tables, with application-level filtering ensuring tenants only access their own records. This approach balances resource efficiency with adequate data isolation [7][9].

Status Workflow Design: The system implements structured status transitions for both orders and payments. Order status follows a sequential workflow: DRAFT (initial creation) → AWAITING_PAYMENT → PROCESSING → READY_TO_SHIP → SHIPPED → COMPLETED, with CANCELLED as a terminal state accessible from early stages. Payment status progresses through: UNPAID → DOWN_PAYMENT_RECEIVED → INSTALLMENT → FULLY_PAID. These status transitions are enforced at the application layer through validation rules in the Orders and Payments modules, ensuring data consistency and proper workflow progression.

3.1.4 System Implementation

The system architecture follows a three-tier pattern: (1) **Presentation Layer** using React application hosted on Vercel with responsive design; (2) **Business Logic Layer** using NestJS application on Railway implementing modular service architecture with dedicated modules for authentication, users, orders, payments, stock, and reporting; and (3) **Data Layer** using PostgreSQL database with Supabase Storage for file uploads.

User interface screenshots are presented in Figures 5-10, demonstrating the key functional interfaces of the SIM4LON system.

![INSERT FIGURE: Login Page Screenshot]

Figure 5. Login Page Interface

![INSERT FIGURE: Dashboard Screenshot]

Figure 6. Admin/Operator Dashboard

![INSERT FIGURE: Order Management Screenshot]

Figure 7. Order Management Interface

![INSERT FIGURE: Create Order Form Screenshot]

Figure 8. Create Order Form

![INSERT FIGURE: Payment Management Screenshot]

Figure 9. Payment Recording Interface

![INSERT FIGURE: Stock Management Screenshot]

Figure 10. Stock Management Interface

3.1.5 System Testing

Black-box testing was conducted across all functional modules. Table 2 presents sample test cases, and Table 3 summarizes the testing results.

Table 2. Sample Black-Box Test Cases

ID	Module	Test Case	Input	Expected Output	Result
TC01	Auth	Valid Login	Correct credentials	Dashboard redirect	Pass
TC02	Auth	Invalid Login	Wrong password	Error message	Pass
TC03	Auth	Session Expiry	Expired JWT	Auto logout	Pass
TC04	Auth	Single Session	New device login	Previous session terminated	Pass
TC05	Order	Create Order	Valid order data	Order created	Pass
TC06	Order	Empty Order	No items	Validation error	Pass
TC07	Order	Status Update	New status	Status changed	Pass
TC08	Order	Driver Assignment	Select driver	Driver assigned	Pass
TC09	Payment	Record DP	Down payment	Payment recorded	Pass

ID	Module	Test Case	Input	Expected Output	Result
TC10	Stock	Record Incoming	Stock addition	Balance updated	Pass

Table 3. Testing Results Summary

Category	Test Cases	Passed	Failed	Success Rate
Authentication	5	5	0	100%
Order Management	8	8	0	100%
Payment Processing	5	5	0	100%
Stock Management	6	6	0	100%
Reporting	4	4	0	100%
Total	**28**	**28**	**0**	**100%**

3.2 Analysis of Findings

The testing results demonstrate that SIM4LON successfully meets all defined functional requirements with a 100% test case pass rate. The system effectively addresses the operational challenges identified during the requirements analysis phase. Order processing becomes streamlined through the status-tracking mechanism, allowing real-time visibility of delivery progress. Payment integration eliminates the need for separate cash book records, ensuring accurate receivables tracking.

The multi-tenant implementation using shared database with shared schema proves effective for this use case. Data isolation through `pangkalan_id` filtering prevents unauthorized access between retail outlets while maintaining a single database instance for operational efficiency. This finding aligns with recommendations from contemporary multi-tenant architecture research [7][9][15].

The technology stack selection of React, NestJS, and PostgreSQL demonstrates suitability for developing scalable web applications. React's component-based architecture enables efficient UI development, NestJS provides structured backend organization through its modular design, and PostgreSQL's ACID compliance ensures transactional data integrity critical for financial record-keeping.

Comparing with previous LPG distribution systems [3][4][5][6], the SIM4LON implementation provides several advancements: (1) modern JavaScript/TypeScript stack replacing legacy PHP implementations; (2) multi-tenant architecture enabling single-deployment multi-client support; (3) cloud-native deployment eliminating on-premise infrastructure requirements; and (4) comprehensive UML documentation facilitating maintainability.

3.3 Implications of the Results

Practical Implications: The system provides tangible benefits for LPG agent operations:

- Reduced order processing time through automated workflows
- Improved stock accuracy through real-time tracking
- Faster report generation (instant vs. 3-5 days manually)
- Enhanced coordination through driver assignment features
- Accessible from any device via web browser

Theoretical Implications: This research contributes to the body of knowledge regarding:

- Application of multi-tenant architecture in distribution management domains
- Integration of modern JavaScript frameworks for enterprise applications
- Implementation patterns for cloud-deployed business systems

Stakeholder Benefits:

- LPG Agents: Operational efficiency and data accuracy improvements
- Retail Outlets (Pangkalan): Real-time order and payment visibility
- Researchers: Reference implementation for similar distribution systems

3.4 Limitations of the Study

Several limitations are acknowledged in this research:

1. **Testing Scope:** Only black-box functional testing was performed. Unit tests, integration tests, and load testing were not conducted, limiting confidence in code quality and performance under high load.

2. **Security Assessment:** Advanced security measures such as rate limiting, penetration testing, and security auditing were not implemented or verified.

3. **User Evaluation:** Formal usability testing with end-users was not conducted. User satisfaction and adoption factors remain unmeasured.

4. **Scale Validation:** The system was not tested with actual production data volumes. Performance with 100+ concurrent retail outlets remains theoretical.

5. **Offline Capability:** The system requires continuous internet connectivity. Offline mode for low-connectivity areas is not supported.

Future research should address these limitations through comprehensive security auditing, usability studies, performance benchmarking with realistic data volumes, and progressive web application features for offline functionality.

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4. Conclusion

This research successfully designed and implemented a web-based LPG distribution management information system with multi-tenant architecture. The main contributions are:

1. **System Design:** A comprehensive UML design was created comprising 18 use cases covering order management, payment processing, stock tracking, and reporting functions. The multi-tenant architecture was implemented using shared database, shared schema pattern with `pangkalan_id` as tenant identifier.

2. **System Implementation:** The SIM4LON system was successfully developed using React 18 for frontend, NestJS 10 for backend API, Prisma ORM for database access, and PostgreSQL 15 for data storage. The system was deployed to cloud infrastructure using Vercel (frontend) and Railway (backend and database).

3. **System Validation:** Black-box testing with 28 test cases covering all functional requirements resulted in 100% pass rate, confirming that the system meets its specified requirements.

The research demonstrates that modern web technologies combined with multi-tenant architecture provide an effective solution for addressing manual process inefficiencies in LPG distribution businesses. The cloud-deployed system offers accessibility advantages while the multi-tenant design enables scalability for serving multiple agents with isolated data.

Recommendations for Future Development:

- Implement comprehensive security features including rate limiting and audit logging
- Develop mobile native applications for enhanced user experience
- Integrate with Pertamina's official distribution tracking systems
- Add offline capability using progressive web application technologies
- Conduct formal usability studies to optimize user interface design

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Declarations

Author contribution: Luthfi Alfaridz conducted the research, developed the system, and wrote the manuscript. Siti Sarah supervised the research and reviewed the manuscript.

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Conflict of interest: The authors declare no conflict of interest.

Additional information: No additional information is available for this paper.

Data and Software Availability Statements

The SIM4LON system is deployed and accessible at:

- Frontend: https://sim4lon.vercel.app
- Backend API: Hosted on Railway platform

Source code is available in a private repository. Access may be granted upon reasonable request to the corresponding author.

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FIGURE PLACEHOLDERS GUIDE

The following 10 figures need to be inserted before submission:

Figure	Type	Description	Section
1	Flowchart	Waterfall Methodology Stages	2.5
2	BPMN	Current System Business Process	3.1.1
3	UML	Use Case Diagram	3.1.3
4	UML	Entity Relationship Diagram	3.1.3
5	Screenshot	Login Page	3.1.4
6	Screenshot	Dashboard	3.1.4
7	Screenshot	Order Management	3.1.4
8	Screenshot	Create Order Form	3.1.4
9	Screenshot	Payment Recording	3.1.4
10	Screenshot	Stock Management	3.1.4

Instructions:

1. Replace each `![INSERT FIGURE: ...]` placeholder with the actual image
2. Ensure all images are high resolution (minimum 300 DPI for print)
3. Use consistent styling across all diagrams
4. Screenshots should use realistic sample data (avoid "test123" or placeholder text)